Test Plan Document

Web-Based ML Model Evaluator

1. Purpose and Scope

This test plan defines the verification approach for the Web-Based ML Model Evaluator. It maps test cases directly to the FR and NFR identifiers in the companion SRS. The plan covers account access, dataset management, preparation, model training, evaluation, comparison, exports, run history, API behaviour, security, usability, and reliability.

Test execution has not been represented as completed in this document. The Actual Result and Test Result columns are intentionally left blank for the team to fill in during execution.

2. Test Objectives

- Verify each functional requirement through positive, negative, integration, or end-to-end tests.
- Verify important quality requirements, especially authorization, user data isolation, data integrity, and error handling.
- Confirm that invalid data and configuration are rejected safely and consistently.
- Maintain traceability from SRS requirement IDs to test case IDs and outcomes.

3. Test Scope

3.1 In Scope

- Registration, login, logout, and protected-resource access.
- CSV upload, validation, preview, statistics, metadata, and dataset replacement.
- Preprocessing, target selection, train-test split, algorithm selection, and training.
- Metrics, evaluation visualizations, model comparison, downloads, and run history.
- API response behaviour, user isolation, reproducibility metadata, browser UI, and failure handling.

3.2 Out of Scope

- Testing algorithms or ML libraries independently of their integration with this application.
- Production-scale penetration testing or formal certification.
- Formats, algorithms, or export types not implemented in the delivered version.

4. Test Approach

Test type	Purpose
Unit Testing (UT)	Test individual validation functions, service methods, metrics calculations, and configuration checks.
Integration Testing (IT)	Test interactions among UI/API, backend, database, file storage, and ML components.
System Testing (ST)	Test complete user workflows and cross-cutting requirements from the user's perspective.
Non-functional Testing (NFT)	Test performance, security, compatibility, reliability, and usability where applicable.

5. Test Case ID Convention

Test Case ID format: <Requirement-ID>-<Test-Type>-<Sequence>, for example FR-2.1-UT-1 or NFR-4-ST-1.

- Requirement-ID: exact FR or NFR identifier defined in the SRS.

- Test-Type: UT = Unit Test, IT = Integration Test, ST = System Test, NFT = Non-Functional Test.
- Sequence: running number for the selected requirement and test type.

6. Test Environment and Test Data

- Run the application in the project's configured development/test environment with backend, database, storage, and ML dependencies available.
- Use dedicated test accounts (User A and User B) to verify user isolation.
- Prepare valid CSV data, an empty CSV, a malformed CSV, a file with missing values, a dataset with numeric and categorical columns, and a dataset with a usable target column.
- Use small deterministic datasets for metric and reproducibility checks; use a larger within-limit dataset for performance tests.
- Do not use real personal or confidential data in test fixtures.

7. Test Execution Summary

The summary below is a planned test inventory, not a claim that tests have already passed.

Requirement ID	Test cases planned	Coverage area
FR-1.1	2	Register with valid required account fields
FR-1.2	2	Authenticate with valid credentials
FR-1.3	1	Logout terminates the active session
FR-1.4	2	Unauthenticated user cannot access protected pages or APIs
FR-2.1	2	Upload a valid CSV dataset
FR-2.2	2	Reject an empty or malformed CSV
FR-2.3	1	Preview displays correct headers and sample rows
FR-2.4	1	Dataset statistics match a known dataset
FR-2.5	2	Replacing a dataset updates the current user's selected dataset
FR-3.1	2	Apply a supported preprocessing option to valid data
FR-3.2	2	Select a valid target column
FR-3.3	2	Accept a valid train-test split
FR-4.1	2	Display supported algorithms and accept a compatible selection
FR-4.2	2	Complete training for a valid configuration
FR-4.3	3	Calculate applicable metrics for classification
FR-4.4	2	Compare two completed runs
FR-5.4	2	Download a supported model artifact or report
FR-6.1	2	Show a user's prior runs and configuration
FR-6.2	2	API returns a successful structured response for a valid request
FR-6.3	1	Show actionable errors for a failed operation
NFR-1	1	Measure response time for common preview/list actions
NFR-10	1	Verify separation of major application components
NFR-11	1	Verify upload/resource limits are enforced
NFR-12	1	Verify run metadata and artifacts map to the correct run
NFR-13	1	Ensure errors do not reveal sensitive internals
NFR-14	1	Deploy from documented setup instructions
NFR-2	1	Show progress/status for a long-running training job
NFR-3	1	Verify passwords are not stored in plaintext
NFR-4	1	Enforce authorization on direct backend requests
NFR-5	1	Verify isolation across datasets, runs, and artifacts

NFR-6	1	Preserve completed results after a later operation fails
NFR-7	1	Retain run configuration needed to explain a result
NFR-8	1	Verify the primary workflow is understandable in the desktop UI
NFR-9	1	Check core workflows in supported browsers
TOTAL	**51**	Planned test cases; execute and record actual outcomes

8. Detailed Test Cases

For each case, record Actual Result and Test Result after execution. Use Pass, Fail, Blocked, or Not Run as appropriate.

FR-1.1

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-1.1-UT-1	3.1 User Access	Register with valid required account fields	Registration page is available.	1. Open registration. 2. Enter valid unique account details. 3. Submit.	Unique username/email; valid password	Account is created and success feedback is shown.		
FR-1.1-UT-2	3.1 User Access	Reject registration with missing or invalid required fields	Registration page is available.	1. Leave a required field blank or enter invalid data. 2. Submit.	Blank required field / invalid email	Validation is shown and no account is created.		

FR-1.2

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-1.2-UT-1	3.1 User Access	Authenticate with valid credentials	A registered test account exists.	1. Open login. 2. Enter valid credentials. 3. Submit.	Valid test username and password	User is authenticated and redirected to the application.		
FR-1.2-UT-2	3.1 User Access	Reject invalid credentials	A registered test account exists.	1. Enter incorrect password. 2. Submit.	Correct username; wrong password	Login is rejected without exposing sensitive account details.		

FR-1.3

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-1.3-IT-1	3.1 User Access	Logout terminates the active session	User is logged in.	1. Select Logout. 2. Navigate back to a protected page.	Authenticated test account	Session ends and protected content requires login again.		

FR-1.4

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-1.4-ST-1	3.1 User Access	Unauthenticated user cannot access protected pages or APIs	User is logged out.	1. Open a protected URL or call a protected endpoint directly.	No session/token	Access is denied or redirected to login; no protected data is returned.		
FR-1.4-ST-2	3.1 User Access	Authenticated user cannot access another user's resource by changing its identifier	User A and User B exist; User B owns a dataset/run.	1. Log in as User A. 2. Request User B's resource identifier.	User B dataset/run ID	The request is denied and no resource details or artifact are disclosed.		

FR-2.1

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-2.1-UT-1	3.2 Dataset Management	Upload a valid CSV dataset	User is authenticated.	1. Open dataset upload. 2. Select a valid CSV. 3. Submit.	CSV with headers and several rows	Upload succeeds and the dataset is associated with the current user.		
FR-2.1-UT-2	3.2 Dataset Management	Reject an unsupported file type	User is authenticated.	1. Select a non-CSV file. 2. Submit.	PDF or executable renamed with a non-CSV extension	Upload is rejected with a clear message.		

FR-2.2

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-2.2-UT-1	3.2 Dataset Management	Reject an empty or malformed CSV	Upload page is available.	1. Upload an empty or malformed CSV. 2. Submit.	Empty file / malformed CSV	Validation fails and unusable data is not accepted.		
FR-2.2-IT-2	3.2 Dataset Management	Validate required tabular structure and report parsing errors	User is authenticated.	1. Upload a CSV with inconsistent rows or no usable columns. 2. Review response.	Malformed rows / missing header	System reports the issue and does not proceed as if the dataset were valid.		

FR-2.3

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-2.3-ST-1	3.2 Dataset Management	Preview displays correct headers and sample rows	A valid dataset is uploaded.	1. Open dataset preview. 2. Compare displayed values with source CSV.	CSV with known headers and rows	Preview shows the correct column names and sample values without unintended row/column shifts.		

FR-2.4

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-2.4-UT-1	3.2 Dataset Management	Dataset statistics match a known dataset	A valid dataset with known dimensions and missing values is uploaded.	1. Open dataset statistics. 2. Compare displayed values to fixture.	10 rows, 3 columns, known missing cells	Displayed dimensions and available missing-value/type information match the fixture.		

FR-2.5

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-2.5-IT-1	3.2 Dataset Management	Replacing a dataset updates the current user's selected dataset	User has already uploaded Dataset A.	1. Upload Dataset B. 2. Open preview and metadata.	Dataset A: 5 rows; Dataset B: 8 rows	Current dataset reflects Dataset B; previous data is not incorrectly shown as current.		
FR-2.5-ST-2	3.2 Dataset Management	Dataset metadata is isolated by owner	User A and User B each upload a dataset.	1. Sign in as each user. 2. Open dataset listing/history.	Different dataset names and row counts	Each user sees only their own datasets unless sharing is explicitly supported.		

FR-3.1

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-3.1-UT-1	3.3 Data Preparation	Apply a supported preprocessing option to valid data	A dataset with missing values or categorical data is uploaded.	1. Select a supported preprocessing option. 2. Apply it.	Dataset with known missing/categorical values	Preprocessing completes and output conforms to the selected option.		
FR-3.1-UT-2	3.3 Data Preparation	Reject incompatible preprocessing configuration	A dataset is uploaded.	1. Select an incompatible or incomplete option. 2. Submit configuration.	Invalid option/parameter combination	System rejects the configuration with a useful message.		

FR-3.2

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-3.2-UT-1	3.3 Data Preparation	Select a valid target column	Dataset has multiple columns.	1. Open target selector. 2. Select an existing target column.	Target column: outcome	Selection is saved and used in the model configuration.		
FR-3.2-UT-2	3.3 Data Preparation	Reject a missing or invalid target selection	Dataset is uploaded.	1. Leave target unset or provide an invalid target identifier. 2. Continue.	No target / nonexistent column	Training cannot start and the user is told to select a valid target.		

FR-3.3

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-3.3-UT-1	3.3 Data Preparation	Accept a valid train-test split	Dataset and target column are selected.	1. Choose supported train/test proportions. 2. Save configuration.	80% train / 20% test	Configuration is accepted and partitions are created with expected proportions, subject to rounding.		
FR-3.3-UT-2	3.3 Data Preparation	Reject an invalid train-test split	Dataset and target column are selected.	1. Enter invalid proportions or values that leave an empty partition. 2. Submit.	Negative value, >100%, or total invalid	Configuration is rejected with a clear validation message.		

FR-4.1

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-4.1-UT-1	3.4 Model Configuration	Display supported algorithms and accept a compatible selection	Dataset and task configuration are available.	1. Open algorithm selection. 2. Select a compatible listed algorithm.	One supported classification or regression algorithm	Selected algorithm is stored in the run configuration.		
FR-4.1-IT-2	3.4 Model Configuration	Prevent incompatible algorithm/task configuration	Dataset/task type is known.	1. Attempt to select or submit an incompatible algorithm/task combination.	Algorithm incompatible with task	System prevents execution or returns a clear validation error.		

FR-4.2

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-4.2-IT-1	3.4 Model Training	Complete training for a valid configuration	Valid dataset, target, split, and algorithm are configured.	1. Start training. 2. Wait for completion.	Small deterministic dataset; supported algorithm	Training completes and produces a run result linked to the selected configuration.		
FR-4.2-IT-2	3.4 Model Training	Handle training failure without corrupting prior runs	A previous completed run exists.	1. Start a run with invalid or deliberately failing configuration. 2. Review status and history.	Invalid configuration or controlled failure	Failure is reported; previous completed run remains intact.		

FR-4.3

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-4.3-UT-1	3.5 Evaluation	Calculate applicable metrics for classification	A classification model has predictions and labels.	1. Evaluate the run. 2. Compare metric values with a hand-calculated fixture.	Known labels and predictions	Applicable classification metrics match the expected calculation within suitable numeric tolerance.		
FR-4.3-UT-2	3.5 Evaluation	Calculate applicable metrics for regression	A regression model has predictions and labels.	1. Evaluate the run. 2. Compare metrics with a hand-calculated fixture.	Known numeric labels and predictions	Applicable regression metrics match expected values within suitable tolerance.		
FR-4.3-ST-3	3.5 Evaluation	Display evaluation visualizations where supported	A completed run has evaluation results.	1. Open results. 2. Inspect applicable charts/plots.	Classification run or regression run	Relevant visualizations render and correspond to the selected run.		

FR-4.4

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-4.4-IT-1	3.5 Comparison	Compare two completed runs	At least two completed runs exist.	1. Select two runs. 2. Open comparison.	Two runs with compatible task/metric types	Comparison shows run identities/configuration and comparable metrics without mixing unrelated results.		
FR-4.4-UT-2	3.5 Comparison	Prevent misleading comparison of incompatible metrics	Runs of incompatible task types or metric sets exist.	1. Select incompatible runs for comparison.	One classification and one regression run	System separates incompatible metrics or explains why a direct comparison is unavailable.		

FR-5.4

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-5.4-IT-1	3.5 Export	Download a supported model artifact or report	A completed run has an available artifact/report.	1. Select download. 2. Open the downloaded file.	Completed run ID	Download succeeds; file is readable and corresponds to the selected run.		
FR-5.4-UT-2	3.5 Export	Handle unavailable artifact or report	A run has no artifact of the requested type.	1. Request the unavailable download.	Missing artifact	System shows a clear error and does not return an unrelated file.		

FR-6.1

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-6.1-IT-1	3.6 Run History	Show a user's prior runs and configuration	A user has completed at least one run.	1. Open run history. 2. Select a prior run.	Run with known algorithm, split, and metrics	History shows the correct run status and available configuration/results.		
FR-6.1-ST-2	3.6 Run History	Keep another user's run history private	Two users have separate runs.	1. Sign in as User A. 2. Inspect history and attempt to request User B's run.	User B run ID	User A cannot see or retrieve User B's run.		

FR-6.2

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-6.2-IT-1	3.6 API	API returns a successful structured response for a valid request	Backend service is running.	1. Call a documented endpoint with valid input. 2. Inspect status and response body.	Valid request payload	Expected success status and documented response structure are returned.		
FR-6.2-IT-2	3.6 API	API returns structured error for invalid input	Backend service is running.	1. Call a documented endpoint with invalid or incomplete input.	Malformed/missing field payload	A suitable error status and structured error response are returned.		

FR-6.3

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
FR-6.3-ST-1	3.6 Error Handling	Show actionable errors for a failed operation	Application is available.	1. Trigger an invalid upload or failed run. 2. Review the UI message.	Malformed CSV or invalid model configuration	Message explains the problem and next action without exposing internal stack traces.		

NFR-1

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-1-NFT-1	4. Non-Functional	Measure response time for common preview/list actions	Test environment and representative within-limit dataset are ready.	1. Run the same preview/list request repeatedly under normal load. 2. Record response times.	Document dataset size and sample count	Response-time measurements are recorded and compared with the project's agreed target.		

NFR-2

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-2-NFT-1	4. Non-Functional	Show progress/status for a long-running training job	A configuration is available that takes longer than a normal request.	1. Start training. 2. Observe UI while it runs.	Long-running but valid training job	The user receives running/progress/status feedback and can tell when the job finishes or fails.		

NFR-3

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-3-ST-1	4. Non-Functional	Verify passwords are not stored in plaintext	A test account can be registered; database access is available to authorized tester.	1. Register a test account. 2. Inspect stored credential representation in test database.	Test account password	Stored value is a secure hash, not the submitted plaintext password.		

NFR-4

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-4-ST-1	4. Non-Functional	Enforce authorization on direct backend requests	User A and User B have separate resources.	1. Authenticate as User A.	User B resource identifier	Backend denies access regardless of frontend		

				2. Call the backend directly for User B's resource.		visibility.		

NFR-5

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-5-ST-1	4. Non-Functional	Verify isolation across datasets, runs, and artifacts	Two users have uploaded data and completed runs.	1. Sign in as each user. 2. Check lists and attempt direct resource access.	Distinct user-owned resources	No cross-user data or artifact is disclosed.		

NFR-6

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-6-IT-1	4. Non-Functional	Preserve completed results after a later operation fails	A completed run exists.	1. Record prior result details. 2. Trigger a separate failing training/export operation. 3. Reopen prior run.	Known completed run	Prior result and metadata remain unchanged and accessible.		

NFR-7

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-7-IT-1	4. Non-Functional	Retain run configuration needed to explain a result	A model run completes.	1. Open run details/history. 2. Verify algorithm, target, split, and supported preprocessing settings.	Known run configuration	Stored details match the settings used for the run, to the extent supported by implementation.		

NFR-8

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-8-ST-1	4. Non-Functional	Verify the primary workflow is understandable in the desktop UI	Application is deployed in test environment.	1. Ask a tester to upload data, configure, train, and view results using UI labels/help only. 2. Record obstacles.	Desktop browser	Workflow steps and validation feedback are understandable; usability observations are recorded.		

NFR-9

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-9-NFT-1	4. Non-Functional	Check core workflows in supported browsers	Supported browser matrix is defined for the release.	1. Execute login, upload, configure, train/results navigation in each target browser.	Current agreed Chrome, Firefox, Edge versions	Core layout and interactions work in each browser; discrepancies are logged.		

NFR-10

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-10-IT-1	4. Non-Functional	Verify separation of major application components	Source repository and test environment are available.	1. Review module boundaries and imports. 2. Run relevant component tests.	Frontend, API/service, persistence, ML components	Responsibilities are separated as documented; no single module unexpectedly owns all concerns.		

NFR-11

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-11-NFT-1	4. Non-Functional	Verify upload/resource limits are enforced	Configured limits are known.	1. Upload a file below the limit. 2. Upload a file above the	Below-limit and above-limit fixtures	Supported input is accepted; over-limit input is rejected or safely		

				limit or a dataset exceeding configured constraints.		constrained with clear feedback.		

NFR-12

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-12-IT-1	4. Non-Functional	Verify run metadata and artifacts map to the correct run	Two runs exist with distinct settings and results.	1. Open each run. 2. Inspect metrics, configuration, and artifact links.	Run A and Run B with different algorithms or targets	Each run displays only its own configuration, metrics, and artifact references.		

NFR-13

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-13-ST-1	4. Non-Functional	Ensure errors do not reveal sensitive internals	A controlled backend error can be triggered in test environment.	1. Trigger the error. 2. Inspect UI and API error output.	Invalid request or controlled exception	User-facing response contains no password, secret, or raw stack trace.		

NFR-14

Test Case ID	Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
NFR-14-ST-1	4. Non-Functional	Deploy from documented setup instructions	Clean test environment is available.	1. Follow setup instructions. 2. Install declared dependencies and start the application.	Clean environment and repository	Application starts using documented steps, or missing/incorrect instructions are recorded as defects.		

9. Entry and Exit Criteria

9.1 Entry Criteria

- A testable build is deployed and its required services are running.
- Test accounts, fixture datasets, and required ML dependencies are available.
- The team has identified the supported algorithms, formats, and browser matrix for the build.

9.2 Exit Criteria

- All planned cases have an execution status recorded.
- Critical access-control, user-isolation, data-loss, and training-workflow defects are resolved or formally documented.
- Failed and blocked tests include notes and reproducible steps.
- Known deviations from the SRS and unsupported behaviour are documented for the evaluator.

10. Defect Reporting

Each defect should include a concise title, affected requirement/test case ID, environment, preconditions, reproduction steps, expected result, actual result, severity, evidence (such as screenshot/log with secrets removed), and assignee. Re-test the failed case after a fix and record the new outcome.

11. Requirement Traceability Notes

Every detailed test case uses the requirement ID from the SRS as its prefix. The inventory includes functional and non-functional checks. Some NFRs (for example maintainability and usability) require review or observation in addition to automated tests. The exact numeric performance target, maximum upload size, supported algorithm list, and supported browser versions must be confirmed by the project team and recorded before final acceptance.

12. Security Test Cases and Cross-Document Traceability

These cases supplement the functional and non-functional tests above. SEC IDs refer to the companion Security Requirements Document. They are planned tests, not completed results; fill Actual Result and Test Result only after execution.

Security test ID convention: <SEC requirement ID>-<test type>-<sequence>. Example: SEC-AZ-2-ST-1. For review-based checks, REV denotes a security configuration/code review.

SEC Requirement	Test Case ID	Linked SRS ID(s)	Description	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
SEC-AUTH-1	SEC-AUTH-1-ST-1	FR-1.4	Anonymous access to protected data is denied	User is logged out.	1. Request a protected dataset, run history, or download endpoint. 2. Inspect response.	No session	Request is denied; no protected content is returned.		
SEC-AUTH-2	SEC-AUTH-2-ST-1	NFR-3	Password storage is not plaintext	Test registration is enabled; authorized test database inspection is available.	1. Register a test user. 2. Inspect stored credential representation.	Known test password	Stored value is a salted password hash and does not equal the plaintext password.		
SEC-AUTH-3	SEC-AUTH-3-ST-1	FR-1.2; NFR-13	Login errors do not reveal account existence	Login endpoint is available.	1. Attempt login with unknown username. 2. Attempt login with known username and wrong password. 3. Compare messages.	Unknown user; incorrect password	Messages do not reveal whether an account exists.		
SEC-AUTH-4	SEC-AUTH-4-NFT-1	NFR-3; NFR-11	Repeated failed requests are throttled if protection is configured	Test environment and rate-limit policy are available.	1. Submit repeated failed login requests. 2. Observe responses and logs.	Repeated invalid credentials	Configured throttling/lockout applies; service remains usable for legitimate requests.		
SEC-SES-1	SEC-SES-1-ST-1	FR-1.2; NFR-3	Session/token handling avoids unnecessary exposure	User can log in; test environment is available.	1. Log in. 2. Inspect URLs, logs, and session transport using approved test tools.	Test account	Credentials/tokens are not exposed in URLs or logs; session handling follows the documented design.		
SEC-SES-2	SEC-SES-2-ST-1	FR-1.3; NFR-3	Expired session requires re-authentication	Session expiry policy is configured or documented.	1. Log in. 2. Wait for or simulate expiry in test environment. 3. Request a protected action.	Expired session	Protected action is denied until the user re-authenticates.		
SEC-SES-3	SEC-SES-3-ST-1	FR-1.3	Logout invalidates the session	User is logged in.	1. Log out. 2. Replay the previous session/token against a protected endpoint.	Previously valid session/token	Old session/token no longer grants access.		
SEC-AZ-1	SEC-AZ-1-ST-1	FR-1.4; NFR-4	Backend enforces authorization independently of UI	A protected endpoint exists.	1. Call endpoint directly without required authorization. 2. Repeat with an authenticated but unauthorized account.	Unauthorized account/session	Backend denies the request and returns no protected data.		
SEC-AZ-2	SEC-AZ-2-ST-1	FR-2.5; FR-6.1; NFR-5	User cannot access another user's datasets or runs	User A and User B each own separate datasets and runs.	1. Sign in as User A. 2. Request User B's dataset/run by identifier. 3. Try its download endpoint.	User B resource ID	All unauthorized access attempts are denied without leaking resource details.		
SEC-DAT-1	SEC-DAT-1-ST-1	NFR-5	Check cross-user exposure in lists, history, and downloads	Two users have distinct resources.	1. Compare each user's dataset list and run history. 2. Try direct resource URLs/IDs.	Distinct user resources	Only the owner's resources are visible or downloadable.		
SEC-DAT-2	SEC-DAT-2-IT-1	FR-2.1; FR-2.2; NFR-11	Reject malformed, unsupported, and	Upload limit and accepted input format	1. Upload valid CSV. 2. Upload empty,	Valid CSV; malformed CSV; unsupported file;	Valid input is accepted; invalid		

			over-limit uploads	are documented.	malformed, non-CSV, and over-limit files.	oversized file	input is rejected with clear feedback and without unsafe processing.		
SEC-DAT-3	SEC-DAT-3-IT-1	FR-4.3; FR-6.1; NFR-6; NFR-12	Failed run does not corrupt a completed run	At least one completed run exists.	1. Record prior run details. 2. Trigger a separate failed operation. 3. Reopen prior run.	Completed run plus failing configuration	Prior run details and artifacts remain unchanged.		
SEC-DAT-4	SEC-DAT-4-IT-1	FR-5.4; NFR-12	Downloaded artifact belongs to selected run	Two runs have distinct IDs/configurations and available artifacts.	1. Select Run A and download its artifact/report. 2. Repeat for Run B. 3. Compare metadata/content.	Run A and Run B	Each download corresponds to the selected run and authorized user.		
SEC-INP-1	SEC-INP-1-UT-1	FR-2.2; FR-3.1; FR-3.2; FR-3.3; FR-4.1; FR-5.4	Backend validates configuration and identifiers	Relevant API endpoints are available.	1. Submit invalid dataset ID, target column, split, preprocessing option, algorithm, and export request directly to the API.	Invalid identifiers and configuration values	Invalid requests are rejected; no unintended training or export occurs.		
SEC-INP-2	SEC-INP-2-REV-1	NFR-10; NFR-13	Review database access for injection resistance	Source code and test environment are available.	1. Review database access patterns. 2. Submit injection-like text to fields that reach database queries.	Strings containing SQL metacharacters	Queries treat user input as data; no unauthorized query behaviour occurs.		
SEC-INP-3	SEC-INP-3-ST-1	NFR-13	User text and errors do not expose executable markup or internals	Text fields and controlled error path are available.	1. Submit script-like text where accepted. 2. Trigger a controlled error. 3. Inspect rendered UI and response.	Script-like string; invalid request	Text is safely rendered and error response omits stack traces, secrets, and internal paths.		
SEC-OPS-1	SEC-OPS-1-REV-1	NFR-3; NFR-13; NFR-14	Check for secrets in repository and frontend bundle	Repository and build artifacts are available to authorized reviewer.	1. Search tracked files and built frontend assets for known test secrets or credential patterns.	Test-only sentinel secret	No real credentials or secrets are committed or exposed in the frontend.		
SEC-OPS-2	SEC-OPS-2-REV-1	NFR-10; NFR-14	Check dependency declarations and review process	Dependency files and repository workflow are available.	1. Verify dependencies are declared. 2. Inspect documented dependency review/scanning process if configured.	Requirements/package manifests	Dependencies are declared and the team can identify the process for reviewing vulnerable packages.		
SEC-OPS-3	SEC-OPS-3-REV-1	NFR-6; NFR-14	Review recovery instructions for persistent data	The application stores persistent data or artifacts.	1. Review backup and restore instructions. 2. If configured, perform a test restore in a safe environment.	Test backup and non-production data	Recovery steps are documented; any unimplemented backup/restore capability is recorded as a limitation.		
SEC-OPS-4	SEC-OPS-4-IT-1	FR-6.3; NFR-13	Security events are logged without sensitive values	Logging is enabled in the test environment.	1. Trigger failed login and denied resource access. 2. Inspect logs.	Test account; unauthorized resource request	Relevant events are recorded without passwords, tokens, or dataset contents.		
SEC-OPS-5	SEC-OPS-5-NFT-1	NFR-3; NFR-14	Verify HTTPS/TLS in deployed environment	A deployed test environment is available.	1. Open the application over HTTPS. 2. Inspect connection/security configuration.	Deployed URL	Credentials and uploaded data use HTTPS/TLS; any local-development HTTP exception is documented.		

12.1 Three-Document Traceability Rules

SRS FR/NFR IDs remain the primary functional and quality requirement identifiers. SEC IDs add security-specific detail. Test cases link to both where applicable. If a SEC requirement is marked proposed in the Security Requirements Document, the team should approve it before treating a failed test as a release-blocking defect.

12.2 Security Test Status

No security test is marked Pass by default. Execute the test, record evidence where appropriate, and fill in Actual Result and Test Result. Code-review checks should record reviewer, date, files/areas reviewed, and findings.

13. Additional Tests for Administrative and Instructor Use Cases

These cases align the test plan with UC-09 and UC-12 to UC-14 in the Use Case Flow document. They are planned tests; record actual outcomes only after execution.

Test Case ID	Requirement Trace	Description	Preconditions	Test Steps	Test Data	Expected Result	Actual Result	Test Result
FR-7.1-ST-1	FR-7.1; SEC-AZ-3	Admin can view registered users	Admin account exists and is authenticated.	1. Open the user-management area. 2. Review the user list.	Several test accounts	The Admin can view the registered accounts allowed by the application.		
FR-7.1-ST-2	FR-7.1; SEC-AZ-3	Admin can perform a supported account action	Admin and a non-admin test account exist.	1. Select a test account. 2. Perform a supported update or deactivation. 3. Recheck the account status.	Dedicated test account	The action is validated, saved, and confirmed; the affected account reflects the change.		
FR-7.1-ST-3	FR-7.1; SEC-AZ-3	Non-admin cannot manage user accounts	A Student/regular user and an Admin exist.	1. Log in as a regular user. 2. Open the user-management route or call its API directly.	Regular-user session	The request is denied and no user-management action is performed.		
FR-7.2-ST-1	FR-7.2; SEC-AZ-4	Admin can view stored dataset metadata	Admin account and stored datasets exist.	1. Open dataset management. 2. Inspect dataset records and metadata.	Test datasets with known owner, rows, and columns	The Admin view shows the supported metadata accurately.		
FR-7.2-ST-2	FR-7.2; SEC-AZ-4	Dataset management preserves ownership and associations	Datasets and runs exist for at least two users.	1. Inspect a dataset record. 2. Perform a supported management action in test data. 3. Check associated run references and owner information.	Two users' test datasets	The operation follows permissions and does not associate a dataset or run with the wrong owner.		
FR-7.2-ST-3	FR-7.2; SEC-AZ-4	Non-admin cannot use administrative dataset management	A regular-user account exists.	1. Log in as a regular user. 2. Access dataset-management UI/API directly.	Regular-user session	Access is denied; no administrative operation occurs.		
FR-7.3-ST-1	FR-7.3; SEC-AZ-6; SEC-AUD-1	Admin can view relevant system activity logs	Admin account and test activity exist.	1. Upload a dataset and run a model. 2. Open the system-log view as Admin. 3. Locate the events.	Known upload and training actions	Available relevant events are displayed with useful event details.		
FR-7.3-ST-2	FR-7.3; SEC-AZ-6; SEC-AUD-1	System logs are read-only and do not expose secrets	Admin can access logs.	1. Inspect available log controls. 2. Review test log entries for sensitive values. 3. Attempt an unsupported modification if an endpoint exists.	Test events; no real secrets	Logs cannot be edited through the UI; passwords, tokens, and dataset contents are not exposed.		
FR-7.3-ST-3	FR-7.3; SEC-AZ-6	Non-admin cannot view administrative logs	A regular-user account exists.	1. Log in as a regular user. 2. Navigate directly to the log route or call the endpoint.	Regular-user session	Access is denied and log content is not returned.		
FR-7.4-ST-1	FR-7.4; FR-5.1; FR-5.2; SEC-AZ-1	Instructor can view available evaluation results	Instructor account and a completed model run available for review exist.	1. Log in as Instructor. 2. Open the available evaluation result.	Completed test run	Evaluation metrics and supported visualizations are displayed read-only.		
FR-7.4-ST-2	FR-7.4; SEC-AZ-5	Instructor cannot modify training inputs or results	Instructor account exists.	1. Attempt to upload/replace a dataset, change a model configuration, start training, or alter a result.	Instructor session	Unauthorized write operations are denied.		
FR-7.4-ST-3	FR-7.4; NFR-5	Instructor cannot access results outside their authorized scope	Instructor and multiple users' runs exist.	1. Request a result not made available to the Instructor by changing its identifier.	Unauthorized run ID	The system denies access and does not disclose the result.		

13.1 Administrative Feature Scope Decision

UC-12 Manage Users, UC-13 Manage Datasets, and UC-14 View System Logs are included in the current document set. Before final submission, confirm these functions are in the planned implementation. If not, update the UML, Use Case Flows, SRS, Security Requirements, and Test Plan consistently to mark them out of scope.